winnt32

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Performs an installation of or upgrade to a product in Windows Server 2003. You can run **winnt32** at the command prompt on a computer running Windows 95, Windows 98, Windows Millennium edition, Windows NT, Windows 2000, Windows XP, or a product in the Windows Server 2003. If you run **winnt32** on a computer running Windows NT version 4.0, you must first apply Service Pack 5 or later.

Syntax

```
winnt32 [/checkupgradeonly] [/cmd: <CommandLine>] [/cmdcons] [/copydir:
{i386|ia64}\<FolderName>] [/copysource: <FolderName>] [/debug[<Level>]:[
<FileName>]] [/dudisable] [/duprepare: <pathName>] [/dushare: <pathName>]
[/emsport:{com1|com2|usebiossettings|off}] [/emsbaudrate: <BaudRate>] [/m:
<FolderName>]  [/makelocalsource] [/noreboot] [/s: <Sourcepath>] [/syspart:
<DriveLetter>] [/tempdrive: <DriveLetter>] [/udf: <ID>[,<UDB_File>]]
[/unattend[<Num>]:[ <AnswerFile>]]
```

Parameters

Expand table

Parameter	Description
/checkupgradeonly	Checks your computer for upgrade compatibility with products in Windows Server 2003. if you use this option with **/unattend**, no user input is required. Otherwise, the results are displayed on the screen, and you can save them under the file name you specify. The default file name is **upgrade.txt** in the systemroot folder.
/cmd	Instructs setup to carry out a specific command before the final phase of setup. This occurs after your computer has restarted and after setup has collected the necessary configuration information, but before setup is complete.
<CommandLine>	Specifies the commandline to be carried out before the final phase of setup.

Parameter	Description
/cmdcons	On an x86-based computer, installs the recovery Console as a startup option. The recovery Console is a command-line interface from which you can perform tasks such as starting and stopping services and accessing the local drive (including drives formatted with NTFS). You can only use the **/cmdcons** option after setup is finished.
/copydir	creates an additional folder within the folder in which the operating system files are installed. for example, for x86 and x64-based computers, you could create a folder called *Private_drivers* within the i386 source folder for your installation, and place driver files in the folder. type **/copydir:i386***Private_drivers* to have setup copy that folder to your newly installed computer, making the new folder location **systemroot***Private_drivers*. - **i386** specifies i386 - **ia64** specifies ia64 You can use **/copydir** to create as many additional folders as you want.
<FolderName>	Specifies the folder that you created to hold modifications for your site.
/copysource	creates a temporary additional folder within the folder in which the operating system files are installed. You can use **/copysource** to create as many additional folders as you want. Unlike the folders **/copydir** creates, **/copysource** folders are deleted after Setup completes.
/debug	creates a debug log at the level specified, for example, **/debug4:Debug.log**. The default log file is **C:\ systemroot\winnt32.log**, and
<level>	Level Values and descriptions - 0: Severe Errors - 1: Errors - 2: Default level. Warnings - 3: Information - 4: detailed information for debugging Each level includes the levels below it.
/dudisable	Prevents Dynamic Update from running. Without Dynamic Update, setup runs only with the original setup files. This option will disable Dynamic Update even if you use an answer file and specify Dynamic Update options in that file.
/duprepare	Carries out preparations on an installation share so that it can be used with Dynamic Update files that you downloaded from the Windows Update Web site. This share can then be used for installing Windows XP for multiple clients.

Parameter	Description
<pathName>	Specifies full path name.
/dushare	Specifies a share on which you previously downloaded Dynamic Update files (updated files for use with Setup) from the Windows Update Web site, and on which you previously ran **/duprepare:**< *pathName*>. When run on a client, specifies that the client installation will make use of the updated files on the share specified in <pathName>.
/emsport	Enables or disables Emergency Management Services during setup and after the server operating system has been installed. With Emergency Management Services, you can remotely manage a server in emergency situations that would typically require a local keyboard, mouse, and monitor, such as when the network is unavailable or the server is not functioning properly. Emergency Management Services has specific hardware requirements, and is available only for products in Windows Server 2003. - **com1** is applicable only for x86-based computers (not Itanium architecture-based computers). - **com2**is applicable only for x86-based computers (not Itanium architecture-based computers). - Default. Uses the setting specified in the BIOS Serial Port Console Redirection (SPCR) table, or, in Itanium architecture-based systems, through the EFI console device path. If you specify **usebiossettings** and there is no SPCR table or appropriate EFI console device path, Emergency Management Serices will not be enabled. - **off** disables Emergency Management Services. You can later enable it by modifying the boot settings.
/emsbaudrate	for x86-based computers, specifies the baud rate for Emergency Management Services. (The option is not applicable for Itanium architecture-based computers.) Must be used with **/emsport:com1** or **/emsport:com2** (otherwise, **/emsbaudrate** is ignored).
<BaudRate>	Specifies baudrate of 9600, 19200, 57600, or 115200. 9600 is the default.
/m	Specifies that setup copies replacement files from an alternate location. Instructs setup to look in the alternate location first, and if files are present, to use them instead of the files from the default location.
/makelocalsource	Instructs setup to copy all installation source files to your local hard disk. Use **/makelocalsource** when installing from a cd to provide installation files when the cd is not available later in the installation.
/noreboot	Instructs setup to not restart the computer after the file copy phase of setup is completed so that you can run another command.
/s	Specifies the source location of the files for your installation. To simultaneously copy files from multiple servers, type the **/s:**<Sourcepath>

Parameter	Description
	option multiple times (up to a maximum of eight). If you type the option multiple times, the first server specified must be available, or setup will fail.
<Sourcepath>	Specifies full source path name.
/syspart	On an x86-based computer, specifies that you can copy setup startup files to a hard disk, mark the disk as active, and then install the disk into another computer. When you start that computer, it automatically starts with the next phase of setup. You must always use the **/tempdrive** parameter with the **/syspart** parameter. You can start **winnt32** with the **/syspart** option on an x86-based computer running Windows NT 4.0, Windows 2000, Windows XP, or a product in Windows Server 2003. If the computer is running Windows NT version 4.0, it requires Service Pack 5 or later. The computer cannot be running Windows 95, Windows 98, or Windows Millennium edition.
<DriveLetter>	Specifies the drive letter.
/tempdrive	directs setup to place temporary files on the specified partition. for a new installation, the server operating system will also be installed on the specified partition. for an upgrade, the **/tempdrive** option affects the placement of temporary files only; the operating system will be upgraded in the partition from which you run **winnt32**.
/udf	Indicates an identifier (<ID>) that setup uses to specify how a Uniqueness Database (UDB) file modifies an answer file (see the **/unattend** option). The UDB overrides values in the answer file, and the identifier determines which values in the UDB file are used. For example, **/udf:RAS_user,Our_company.udb** overrides settings specified for the RAS_user identifier in the Our_company.udb file. If no <UDB_file> is specified, setup prompts the user to insert a disk that contains the **$Unique$.udb** file.
<ID>	Indicates an identifier used to specify how a Uniqueness Database (UDB) file modifies an answer file.
<UDB_file>	Specifies a Uniqueness Database (UDB) file.
/unattend	On an x86-based computer, upgrades your previous version of Windows NT 4.0 Server (with Service Pack 5 or later) or Windows 2000 in unattended setup mode. All user settings are taken from the previous installation, so no user intervention is required during setup.
<num>	Specifies the number of seconds between the time that setup finishes copying the files and when it restarts your computer. You can use <Num> on any computer running Windows 98, Windows Millennium edition,

Parameter	Description
	Windows NT, Windows 2000, Windows XP, or a product in Windows Server 2003 . If the computer is running Windows NT version 4.0, it requires Service Pack 5 or later.
<AnswerFile>	Provides setup with your custom specifications
/?	Displays help at the command prompt.

Remarks

If you are deploying Windows XP on client computers, you can use the version of winnt32.exe that comes with Windows XP. Another way to deploy Windows XP is to use winnt32.msi, which works through Windows Installer, part of the IntelliMirror set of technologies. For more information about client deployments, see the Windows Server 2003 Deployment Kit, which is described in Using the Windows Deployment and Resource Kits.

On an Itanium-based computer, **winnt32** can be run from the Extensible Firmware Interface (EFI) or from Windows Server 2003 Enterprise, Windows Server 2003 R2 Enterprise, Windows Server 2003 R2 Datacenter, or Windows Server 2003 Datacenter. Also, on an Itanium architecture-based computer, **/cmdcons** and **/syspart** are not available, and options relating to upgrades are not available. for more information about hardware compatibility, see Hardware compatibility. for more detailed information about using Dynamic Update and installing multiple clients, see the Windows Server 2003 Deployment Kit, which is described in Using the Windows Deployment and Resource Kits. for information about modifying boot settings, see the Windows Deployment and Resource Kits for Windows Server 2003. For more information, see Using the Windows Deployment and Resource Kits. Using the **/unattend** command-line option to automate setup affirms that you have read and accepted the Microsoft License Agreement for Windows Server 2003. Before using this command-line option to install Windows Server 2003 on behalf of an organization other than your own, you must confirm that the end user (whether an individual, or a single entity) has received, read, and accepted the terms of the Microsoft License Agreement for that product. OEMs may not specify this key on machines being sold to end users.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

winrs

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Windows remote Management using the winrs command enables you to manage and execute programs remotely.

Syntax

```
winrs [/<parameter>[:<value>]] <command>
```

Parameters

Expand table

Parameter	Description
/remote: <endpoint>	Specifies the target endpoint using a NetBIOS name or the standard connection: - <url>: [<transport>://]<target>[:<port>] if not specified, **/r:localhost** is used.
/unencrypted	Specifies that the messages to the remote shell aren't encrypted. This parameter is useful for troubleshooting or when the network traffic is already encrypted using **ipsec**, or when physical security is enforced. By default, the messages are encrypted using Kerberos or NTLM keys. This command-line option is ignored when HTTPS transport is selected.
/username: <username>	Specifies username on command line. If not specified, the tool uses Negotiate authentication or prompt for the name. If **/username** is specified, **/password** must also be specified.
/password: <password>	Specifies password on command line. If **/password** isn't specified but **/username** is, the tool prompts for the password.

Parameter	Description
	If **/password** is specified, **/username** must also be specified.
/timeout:<seconds>	This option is deprecated.
/directory:<path>	Specifies starting directory for remote shell. If not specified, the remote shell starts in the user's home directory defined by the environment variable **%USERPROFILE%**.
/environment:<string>=<value>	Specifies a single environment variable to be set when shell starts, which allows changing default environment for shell. Multiple occurrences of this switch must be used to specify multiple environment variables.
/noecho	Specifies that echo should be disabled. noecho may be necessary to ensure that user's answers to remote prompts aren't displayed locally. By default echo is on.
/noprofile	Specifies that the user's profile shouldn't be loaded. By default, the server attempts to load the user profile. If the remote user isn't a local administrator on the target system, then this option is required (the default results in error.)
/allowdelegate	Specifies that the user's credentials can be used to access a remote share, for example, a different machine than the target endpoint.
/compression	Turn on compression. Older installations on remote machines might not support compression. Default setting is off because older installations on remote machines might not support compression.
/usessl	Use an SSL connection when using a remote endpoint. Specifying SSL instead of the transport **https:** uses the default **WinRM** default port.
/?	Displays help at the command prompt.

Remarks

- All command-line options accept either short form or long form. For example, both **/r** and **/remote** are valid.
- To terminate the **/remote** command, the user can type **Ctrl-C** or **Ctrl-break**, which is sent to the remote shell. The second **Ctrl-C** forces termination of **winrs.exe**.
- To manage active remote shells or winrs configuration, use the WinRM tool. The URI alias to manage active shells is **shell/cmd**. The URI alias for winrs configuration is **winrm/config/winrs**.

Examples

```
winrs /r:https://contoso.com command
```

```
winrs /r:contoso.com /usessl command
```

```
winrs /r:myserver command
```

```
winrs /r:http://127.0.0.1 command
```

```
winrs /r:http://169.51.2.101:80 /unencrypted command
```

```
winrs /r:https://[::FFFF:129.144.52.38] command
```

```
winrs /r:http://[1080:0:0:0:8:800:200C:417A]:80 command
```

```
winrs /r:https://contoso.com /t:600 /u:administrator /p:$%fgh7 ipconfig
```

```
winrs /r:myserver /env:path=^%path^%;c:\tools /env:TEMP=d:\temp config.cmd
```

```
```

```
winrs /r:myserver netdom join myserver /domain:testdomain /userd:johns
/passwordd:$%fgh789
```

```
winrs /r:myserver /ad /u:administrator /p:$%fgh7 dir \\anotherserver\share
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

winsat mem

Article • 05/04/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The `winsat mem` command tests system memory bandwidth using a process similar to the large memory-to-memory buffer copies in multimedia processing.

Syntax

CLI

```
winsat mem <parameters>
```

Parameters

Expand table

Parameter	Description
`-up`	Force memory testing with only one thread. The default is to run one thread per physical CPU or core.
`-rn`	Specify that the assessment's threads should run at normal priority. The default is to run at priority 15.
`-nc`	Specify that the assessment should allocate memory and flag it as uncached. This flag means that the processor's caches is bypassed for copy operations. The default is to run in cached space.
`-do <n>`	Specify the distance in bytes, between the end of the source buffer and the beginning of the destination buffer. The default is 64 bytes. The maximum allowable destination offset is 16 MB. Specifying an invalid destination offset results in an error. Note: Zero is a valid value for `<n>`, but negative numbers aren't.
`-mint <n>`	Specify the minimum run time in seconds for the assessment. The default is 2.0. The minimum value is 1.0. The maximum value is 30.0. Note: Specifying a `-mint` value greater than the `-maxt` value when the two parameters are used in combination results in an error.

Parameter	Description
`-maxt <n>`	Specify the maximum run time in seconds for the assessment. The default is 5.0. The minimum value is 1.0. The maximum value is 30.0. If used in combination with the `-mint` parameter, the assessment will begin to do periodic statistical checks of its results after the period of time specified in `-mint`. If the statistical checks pass, then the assessment finishes before the period of time specified in `-maxt` has elapsed. If the assessment runs for the period of time specified in `-maxt` without satisfying the statistical checks, then the assessment finishes at that time and return the results it has collected.
`-buffersize <n>`	Specify the buffer size that the memory copy test should use. Twice this amount is allocated per CPU, which determines the amount of data copied from one buffer to another. The default is 16 MB. This value is rounded to the nearest 4-KB boundary. The maximum value is 32 MB. The minimum value is 4 KB. Specifying an invalid buffer size results in an error.
`-v`	Send verbose output to STDOUT, including status and progress information. Any errors are also be written to the command window.
`-xml <file name>`	Save the output of the assessment as the specified XML file. If the specified file exists, it will be overwritten.
`-idiskinfo`	Save information about physical volumes and logical disks as part of the `<SystemConfig>` section in the XML output.
`-iguid`	Create a globally unique identifier (GUID) in the XML output file.
`-note <note text>`	Add the note text to the `<note text>` section in the XML output file.
`-icn`	Include the local computer name in the XML output file.
`-eef`	Enumerate extra system information in the XML output file.

Examples

- To run the assessment for a minimum of 4 seconds and no longer than 12 seconds, using a 32-MB buffer size and saving the results in XML format to the file `memtest.xml`:

CLI

```
winsat mem -mint 4.0 -maxt 12.0 -buffersize 32MB -xml memtest.xml
```

Remarks

- Minimum requirement to use the `winsat mem` command is membership in the local Administrators group (or equivalent). `Winsat mem` must be executed from an elevated command prompt window.
- To open an elevated command prompt window, select **Start**, select **Accessories**, right-select **Command Prompt**, and select **Run as administrator**.

Feedback

Was this page helpful? Yes No

winsat mfmedia

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Measures the performance of video decoding (playback) using the Media Foundation framework.

Syntax

```
winsat mfmedia <parameters>
```

Parameters

Expand table

Parameters	Description
-input <file name>	Required: Specify the file containing the video clip to be played or encoded. The file can be in any format that can be rendered by Media Foundation.
-dumpgraph	Specify that the filter graph should be saved to a GraphEdit-compatible file before the assessment starts.
-ns	Specify that the filter graph should run at the normal playback speed of the input file. By default, the filter graph runs as fast as possible, ignoring presentation times.
-play	Run the assessment in decode mode and play any supplied audio content in the file specified in **-input** using the default DirectSound device. By default, audio playback is disabled.
-nopmp	Do not make use of the Media Foundation Protected Media Pipeline (MFPMP) process during the assessment.
-pmp	Always make use of the MFPMP process during the assessment. Note: If **-pmp** or **-nopmp** is not specified, MFPMP will be used only when necessary.
-v	Send verbose output to STDOUT, including status and progress information. Any errors will also be written to the command window.

Parameters	Description
-xml <file name>	Save the output of the assessment as the specified XML file. If the specified file exists, it will be overwritten.
-idiskinfo	Save information about physical volumes and logical disks as part of the **<SystemConfig>** section in the XML output.
-iguid	Create a globally unique identifier (GUID) in the XML output file.
-note note text	Add the note text to the **<note>** section in the XML output file.
-icn	Include the local computer name in the XML output file.
-eef	Enumerate extra system information in the XML output file.

Examples

- To runs the assessment with the input file that is used during a **winsat formal** assessment, without employing the Media Foundation Protected Media Pipeline (MFPMP), on a computer where c:\windows is the location of the Windows folder.

```
winsat mfmedia -input c:\windows\performance\winsat\winsat.wmv -nopmp
```

Remarks

- Membership in the local Administrators group, or equivalent, is the minimum required to use **winsat**. The command must be executed from an elevated command prompt window.
- To open an elevated command prompt window, click **Start**, click **Accessories**, right-click **Command Prompt**, and click **Run as administrator**.

Related links

Feedback

Was this page helpful? Yes No

wmic

06/02/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Displays WMI information inside an interactive command shell.

> **ⓘ Important**
>
> The WMI command-line (WMIC) utility is deprecated as of Windows 10, version 21H1, and as of the 21H1 semi-annual channel release of Windows Server. This utility is superseded by Windows PowerShell for WMI (see Chapter 7—Working with WMI). This deprecation applies only to the WMI command-line (WMIC) utility; Windows Management Instrumentation (WMI) itself is not affected. Also see Windows 10 features we're no longer developing.

Syntax

Windows Command Prompt

```
wmic </parameter>
```

Sub-commands

The following sub-commands are available:

Expand table

Sub-command	Description
logicaldisk	Displays properties of all the logical disks on this computer.
class	Escapes from the default alias mode of WMIC to access classes in the WMI schema directly.
path	Escapes from the default alias mode of WMIC to access instances in the WMI schema directly.
context	Displays the current values of all global switches.

Sub-command	Description
[quit | exit]	Exits the WMIC command shell.

Examples

To display the current values of all global switches, type:

Windows Command Prompt

```
wmic context
```

Output similar to the following displays:

Output

```
NAMESPACE    : root\cimv2
ROLE         : root\cli
NODE(S)      : BOBENTERPRISE
IMPLEVEL     : IMPERSONATE
[AUTHORITY   : N/A]
AUTHLEVEL    : PKTPRIVACY
LOCALE       : ms_409
PRIVILEGES   : ENABLE
TRACE        : OFF
RECORD       : N/A
INTERACTIVE  : OFF
FAILFAST     : OFF
OUTPUT       : STDOUT
APPEND       : STDOUT
USER         : N/A
AGGREGATE    : ON
```

To display all properties of all logical disks on the computer, type:

Windows Command Prompt

```
wmic logicaldisk
```

Output similar to the following displays:

Output

```
Access                    Description        InstallDate
PowerManagementSupported   StatusInfo
Availability              DeviceID           LastErrorCode
```

```
ProviderName                SupportsDiskQuotas
BlockSize                   DriveType           MaximumComponentLength        Purpose
SupportsFileBasedCompression
Caption                     ErrorCleared        MediaType
QuotasDisabled               SystemCreationClassName
Compressed                  ErrorDescription    Name
QuotasIncomplete             SystemName
ConfigManagerErrorCode      ErrorMethodology    NumberOfBlocks
QuotasRebuilding             VolumeDirty
ConfigManagerUserConfig     FileSystem          PNPDeviceID                   Size
VolumeName
CreationClassName           FreeSpace           PowerManagementCapabilities   Status
VolumeSerialNumber

0   C:   FALSE   Win32_LogicalDisk   Local Fixed Disk   C:   3   NTFS
543237451776   255   12   C:
1021771247616   FALSE   TRUE   Win32_ComputerSystem   <TheDeviceName>
<TheVolumeName>   0011AABB
```

To get the name of the logical disks, type:

Windows Command Prompt

```
wmic logicaldisk get name
```

Output similar to the following displays:

Output

```
Name
C:
```

To change the language ID used by the command line to English (locale ID 409), type:

Windows Command Prompt

```
wmic /locale:ms_409
```

Related links

- Command-Line Syntax Key

wscript

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

Windows Script Host provides an environment in which users can execute scripts in various languages that use various object models to perform tasks.

Syntax

```
wscript [<scriptname>] [/b] [/d] [/e:<engine>] [{/h:cscript|/h:wscript}]
[/i] [/job:<identifier>] [{/logo|/nologo}] [/s] [/t:<number>] [/x] [/?]
[<ScriptArguments>]
```

Parameters

Expand table

Parameter	Description
`scriptname`	Specifies the path and file name of the script file.
`/b`	Specifies batch mode, which doesn't display alerts, scripting errors, or input prompts. The /b parameter is the opposite of **/i**.
`/d`	Starts the debugger.
`/e`	Specifies the engine that is used to run the script. This parameter lets you run scripts that use a custom file name extension. Without the /e parameter, you can only run scripts that use registered file name extensions. For example, if you try to run this command: `cscript test.admin` You receive this error message: Input Error: There's no script engine for file extension `.admin`. One advantage of using nonstandard file name extensions is that it guards against accidentally double-selecting a script and running something you didn't actually want to run. Using this extension doesn't create a permanent association between the

Parameter	Description
	`.admin` file name extension and VBScript. Each time you run a script that uses a `.admin` file name extension, you need to use the /e parameter.
`/h:cscript`	Registers **cscript.exe** as the default script host for running scripts.
`/h:wscript`	Registers **wscript.exe** as the default script host for running scripts. This behavior is the default when the **/h** option is omitted.
`/i`	Specifies interactive mode, which displays alerts, scripting errors, and input prompts. The /i parameter is the default and the opposite of **/b**.
`/job:\<identifier>`	Runs the job identified by *identifier* in a **.wsf** script file.
`/logo`	Specifies that the Windows Script Host banner is displayed in the console before the script runs. The /logo parameter is the default and the opposite of **/nologo**.
`/nologo`	Specifies that the Windows Script Host banner isn't displayed before the script runs. The /nologo parameter is the opposite of **/logo**.
`/s`	Saves the current command prompt options for the current user.
`/t:\<number>`	Specifies the maximum time the script can run (in seconds). You can specify up to 32,767 seconds. The default is no time limit.
`/x`	Starts the script in the debugger.
`ScriptArguments`	Specifies the arguments passed to the script. Each script argument must be preceded with a slash (/).
`/?`	Displays Help at the command prompt.

Remarks

- Performing this task doesn't require you to have administrative credentials. Consider performing this task as a user without administrative credentials as a security best practice.
- To open a command prompt, on the **Start** screen, type **cmd**, and then select **command prompt**.
- Each parameter is optional. However you can't specify script arguments without specifying a script. If you don't specify a script or any script arguments, **wscript.exe** displays the **Windows Script Host Settings** dialog box. Use the dialog box to set

global scripting properties for all scripts that **wscript.exe** runs on the local computer.

- The **/t** parameter prevents excessive running of scripts by setting a timer. When the time exceeds the specified value, **wscript** interrupts the script engine and ends the process.
- Windows script files usually have one of the following file name extensions: **.wsf**, **.vbs**, **.js**.
- If you double-select a script file with an extension that has no association, the **Open With** dialog box appears. Select **wscript** or **cscript**, and then select **Always use this program to open this file type**. This registers **wscript.exe** or **cscript.exe** as the default script host for files of this file type.
- You can set properties for individual scripts. For more information about Windows Script Host, see Windows Script Host overview.
- Windows Script Host can use **.wsf** script files. Each **.wsf** file can use multiple scripting engines and perform multiple jobs.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

xcopy

Article • 05/28/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies files and directories, including subdirectories.

For examples of how to use this command, see Examples.

Syntax

```
xcopy <Source> [<Destination>] [/w] [/p] [/c] [/v] [/q] [/f] [/l] [/g] [/d
[:MM-DD-YYYY]] [/u] [/i] [/s [/e]] [/t] [/k] [/r] [/h] [{/a | /m}] [/n] [/o]
[/x] [/exclude:FileName1[+[FileName2]][+[FileName3]]] [{/y | /-y}] [/z] [/b]
[/j] [/compress]
```

Parameters

Expand table

Parameter	Description
<Source>	Required. Specifies the location and names of the files you want to copy. This parameter must include either a drive or a path.
[<Destination>]	Specifies the destination of the files you want to copy. This parameter can include a drive letter and colon, a directory name, a file name, or a combination of these.
/w	Displays the following message and waits for your response before starting to copy files: **Press any key to begin copying file(s)**
/p	Prompts you to confirm whether you want to create each destination file.
/c	Ignores errors.
/v	Verifies each file as it is written to the destination file to make sure that the destination files are identical to the source files.

Parameter	Description
/q	Suppresses the display of `xcopy` messages.
/f	Displays source and destination file names while copying.
/l	Generates a list of files that are to be copied, but doesn't actively copy the files.
/g	Creates decrypted *destination* files when the destination doesn't support encryption.
/d [:MM-DD-YYYY]	Copies source files changed on or after the specified date only. If you don't include a *MM-DD-YYYY* value, `xcopy` copies all *source* files that are newer than existing *destination* files. This command-line option allows you to update files that have changed.
/u	Copies files from *source* that exist on *destination* only.
/i	If *source* is a directory or contains wildcards and *destination* doesn't exist, `xcopy` assumes *destination* specifies a directory name and creates a new directory. Then, `xcopy` copies all specified files into the new directory. By default, `xcopy` prompts you to specify whether *destination* is a file or a directory.
/s	Copies directories and subdirectories, unless they're empty. If you omit **/s**, `xcopy` works within a single directory.
/e	Copies all subdirectories, even if they're empty. Use **/e** with the **/s** and **/t** command-line options.
/t	Copies the subdirectory structure (that is, the tree) only, not files. To copy empty directories, you must include the **/e** command-line option.
/k	Copies files and retains the read-only attribute on *destination* files if present on the *source* files. By default, `xcopy` removes the read-only attribute.
/r	Copies read-only files.
/h	Copies files with hidden and system file attributes. By default, `xcopy` doesn't copy hidden or system files
/a	Copies only *source* files that have their archive file attributes set. **/a** doesn't modify the archive file attribute of the source file. For information about how to set the archive file attribute by using **attrib**, see Related links.
/m	Copies *source* files that have their archive file attributes set. Unlike **/a**, **/m** turns off archive file attributes in the files that are specified in the

Parameter	Description
	source. For information about how to set the archive file attribute by using **attrib**, see Related links.
/n	Creates copies by using the NTFS short file or directory names. **/n** is required when you copy files or directories from an NTFS volume to a FAT volume or when the FAT file system naming convention (that is, 8.3 characters) is required on the *destination* file system. The *destination* file system can be FAT or NTFS.
/o	Copies file ownership and discretionary access control list (DACL) information.
/x	Copies file audit settings and system access control list (SACL) information (implies **/o**).
/exclude:FileName1[+ [FileName2]][+ [FileName3]( )]	Specifies a list of files. At least one file must be specified. Each file contains search strings with each string on a separate line in the file. When any of the strings match any part of the absolute path of the file to be copied, that file is excluded from being copied. For example, specifying the string **obj** will exclude all files underneath the directory **obj** or all files with the **.obj** extension.
/y	Suppresses prompting to confirm that you want to overwrite an existing destination file.
/-y	Prompts to confirm that you want to overwrite an existing destination file.
/z	Copies over a network in restartable mode.
/b	Copies the symbolic link instead of the files. This parameter was introduced in Windows Vista®.
/j	Copies files without buffering. Recommended for very large files. This parameter was added in Windows Server 2008 R2.
/compress	Request network compression during file transfer where applicable.
/[-]sparse	Enables or disables retaining the sparse state of files during copy process. If both parameters are specified, **/-sparse** overrides **/sparse**.
/noclone	Doesn't attempt block cloning as an optimization.
/?	Displays help at the command prompt.

Remarks

- Using **/z**

If you lose your connection during the copy phase (for example, if the server going offline severs the connection), it resumes after you reestablish the connection. **/z** also displays the percentage of the copy operation completed for each file.

- Using **/y** in the COPYCMD environment variable.

  You can use **/y** in the COPYCMD environment variable. You can override this command by using **/-y** on the command line. By default, you're prompted to overwrite.

- Copying encrypted files

  Copying encrypted files to a volume that doesn't support EFS results in an error. Decrypt the files first or copy the files to a volume that does support EFS.

- Appending files

  To append files, specify a single file for destination, but multiple files for source (that is, by using wildcards or file1+file2+file3 format).

- Default value for *destination*

  If you omit *destination*, the `xcopy` command copies the files to the current directory.

- Specifying whether *destination* is a file or directory

  If *destination* doesn't contain an existing directory and doesn't end with a backslash (), the following message appears:

  ```
  Does <Destination> specify a file name or directory name on the
  target(F = file, D = directory)?
  ```

  Press F if you want the file or files to be copied to a file. Press D if you want the file or files to be copied to a directory.

  You can suppress this message by using the **/i** command-line option, which causes `xcopy` to assume that the destination is a directory if the source is more than one file or a directory.

- Using the `xcopy` command to set archive attribute for *destination* files

  The `xcopy` command creates files with the archive attribute set, whether or not this attribute was set in the source file. For more information about file attributes and

attrib, see Related links.

- Comparing `xcopy` and `diskcopy`

  If you have a disk that contains files in subdirectories and you want to copy it to a disk that has a different format, use the `xcopy` command instead of `diskcopy`. Because the `diskcopy` command copies disks track by track, your source and destination disks must have the same format. The `xcopy` command doesn't have this requirement. Use `xcopy` unless you need a complete disk image copy.

- Insufficient memory error

  An "insufficient memory" error may occur if running `xcopy` to copy a file or folder whose filename path is greater than 255 characters.

- Exit codes for `xcopy`

  To process exit codes returned by `xcopy`, use the **ErrorLevel** parameter on the **if** command line in a batch program. For an example of a batch program that processes exit codes using **if**, see Related links. The following table lists each exit code and a description.

Expand table

Exit code	Description
0	Files were copied without error.
1	No files were found to copy.
2	The user pressed CTRL+C to terminate `xcopy`.
4	Initialization error occurred. There isn't enough memory or disk space, or you entered an invalid drive name or invalid syntax on the command line.
5	Disk write error occurred.

Examples

1. To copy all the files and subdirectories (including any empty subdirectories) from drive A to drive B, type:

```
```

```
xcopy a: b: /s /e
```

2. To include any system or hidden files in the previous example, add the **/h** command-line option as follows:

```
xcopy a: b: /s /e /h
```

3. To update files in the \Reports directory with the files in the \Rawdata directory that have changed since December 29, 1993, type:

```
xcopy \rawdata \reports /d:12-29-1993
```

4. To update all the files that exist in \Reports in the previous example, regardless of date, type:

```
xcopy \rawdata \reports /u
```

5. To obtain a list of the files to be copied by the previous command (that is, without actually copying the files), type:

```
xcopy \rawdata \reports /d:12-29-1993 /l > xcopy.out
```

The file xcopy.out lists every file that is to be copied.

6. To copy the \Customer directory and all subdirectories to the directory \\Public\Address on network drive H:, retain the read-only attribute, and be prompted when a new file is created on H:, type:

```
xcopy \customer h:\public\address /s /e /k /p
```

7. To issue the previous command, ensure that `xcopy` creates the \Address directory if it doesn't exist, and suppress the message that appears when you create a new directory,

add the **/i** command-line option as follows:

```
xcopy \customer h:\public\address /s /e /k /p /i
```

8. You can create a batch program to perform `xcopy` operations and use the batch **if** command to process the exit code if an error occurs. For example, the following batch program uses replaceable parameters for the `xcopy` source and destination parameters:

```
@echo off
rem COPYIT.BAT transfers all files in all subdirectories of
rem the source drive or directory (%1) to the destination
rem drive or directory (%2)
xcopy %1 %2 /s /e
if errorlevel 4 goto lowmemory
if errorlevel 2 goto abort
if errorlevel 0 goto exit
:lowmemory
echo Insufficient memory to copy files or
echo invalid drive or command-line syntax.
goto exit
:abort
echo You pressed CTRL+C to end the copy operation.
goto exit
:exit
```

To use the preceding batch program to copy all files in the C:\Prgmcode directory and its subdirectories to drive B, type:

```
copyit c:\prgmcode b:
```

The command interpreter substitutes **C:\Prgmcode** for *%1* and **B:** for *%2*, then uses `xcopy` with the **/e** and **/s** command-line options. If `xcopy` encounters an error, the batch program reads the exit code and goes to the label indicated in the appropriate **IF ERRORLEVEL** statement, then displays the appropriate message and exits from the batch program.

9. This example copies all the non-empty directories, plus files with the associated file extension after the asterisk symbol.

```
```

```
xcopy .\toc*.yml ..\..\Copy-To\ /S /Y

rem Output example.
rem  .\d1\toc.yml
rem  .\d1\d12\toc.yml
rem  .\d2\toc.yml
rem  3 File(s) copied
```

In the preceding example, this particular source parameter value **.\toc*.yml** copies the same 3 files even if its two path characters **.** were removed. However, no files would be copied if the asterisk wildcard was removed from the source parameter, making it just **.\toc.yml**.

Related links

- Copy
- Move
- Dir
- Attrib
- Diskcopy
- If
- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No